SCHOOL OF EDUCATION FACULTY WORKSHOP

Humans in the Loop: AI for Learning and Research

October 16, 2026

Megan Alstot, Ph.D., Postdoc for AI & Emerging Technologies for Teaching, ATL
Amy James, Associate Librarian, Online Librarian for Education and Information Literacy, University Libraries

Our session

15 min	20 min	20 min	20 min	10 min
Opening: 4Ds and the Continuum	**Human in the loop for learning**	**Human in the loop for research**	**Work time**	**Debrief**
Megan and Amy	Megan	Amy	Your choice	All of us

One thread runs through both halves: **the human stays in the loop.**

“You didn't ask for AI to arrive at [your] educational doorstep. . . You didn't ask for an unregulated, untested, unreliable, untraceable, and unrelenting technology to upend education… it is clear the genie is out of the bottle… it is not going back in nor is it going to slow down for us to catch up.”

Neither did your students.

Ludwig & Zakrajsek (2026), The Science of Learning Meets AI, p. 3

The human stays in the loop



You



Your students



All of us

You're a human in a learning loop, teaching humans in your learning loops, and every one of us is figuring out how to learn in an AI-integrated loop.

PART 1 · THE LEARNING LOOP

The student does the thinking and decision making the course exists to build.

PART 2 · THE RESEARCH LOOP

The student as researcher judges what the model returns.

The 4Ds of AI Fluency



Delegation

Setting goals and deciding whether, when, and how to engage AI at all.



Description

Describing your goals clearly enough to prompt useful AI behavior, including what you don't want.



Discernment

Accurately assessing the usefulness, quality, and validity of AI outputs and behaviors.



Diligence

Taking responsibility for what we do with AI and how we do it. Responsibility rests with the human.

Dakan, Feller, & Anthropic (2025), CC BY-NC-SA 4.0

The 3As and 4Ds of AI Fluency

Automation: AI executes specific tasks based on human instruction.

Augmentation: Humans and AI collaborate as thinking partners.

Agency: Humans configure AI to independently perform future tasks on their behalf.

Delegation: Setting goals and deciding whether, when, and how to engage AI at all.

Description: Describing your goals clearly enough to prompt useful AI behavior, including what you don't want.

Discernment: Accurately assessing the usefulness, quality, and validity of AI outputs and behaviors.

Diligence: Taking responsibility for what we do with AI and how we do it. Responsibility rests with the human.

Dakan, Feller, & Anthropic (2025), CC BY-NC-SA 4.0

The AI Literacy Continuum

Where does one of your assignments sit today?
Where would you want it to sit?

LEVEL 1

No AI Use

Work completed without AI

WHY

Evidence of independent competence (e.g., domain knowledge)

LEVEL 2

Limited AI Use

AI for narrow, specified tasks

WHY

Lower non-essential barriers (e.g., grammar)

LEVEL 3

Collaborative AI Use

AI as a thought partner throughout

WHY

Engage critical thinking through reflection

LEVEL 4

Integrated AI Use

AI woven in as in professional practice

WHY

Co-create with human-directed interaction

LEVEL 5

AI Innovation

AI used to create novel approaches

WHY

Evidence of AI fluency (discernment and stewardship)

IWU AI Literacy Continuum (Bleistein & Snyder, 2026), adapted from the AIAS (Perkins et al., 2024)

PART 1

Human in the loop for learning

Diagnosing AI's role by stress-testing an assignment

Three kinds of cognitive load

INTRINSIC	The inherent complexity of the content.	Shaped by **the topic or subject**
EXTRANEOUS	Effort spent on how material is presented, not on the content itself.	Shaped by **learning design**
GERMANE	Effort devoted to building understanding.	Shaped by **the learner**

Sweller et al. (1998)

When used well,

AI can fortify learning.

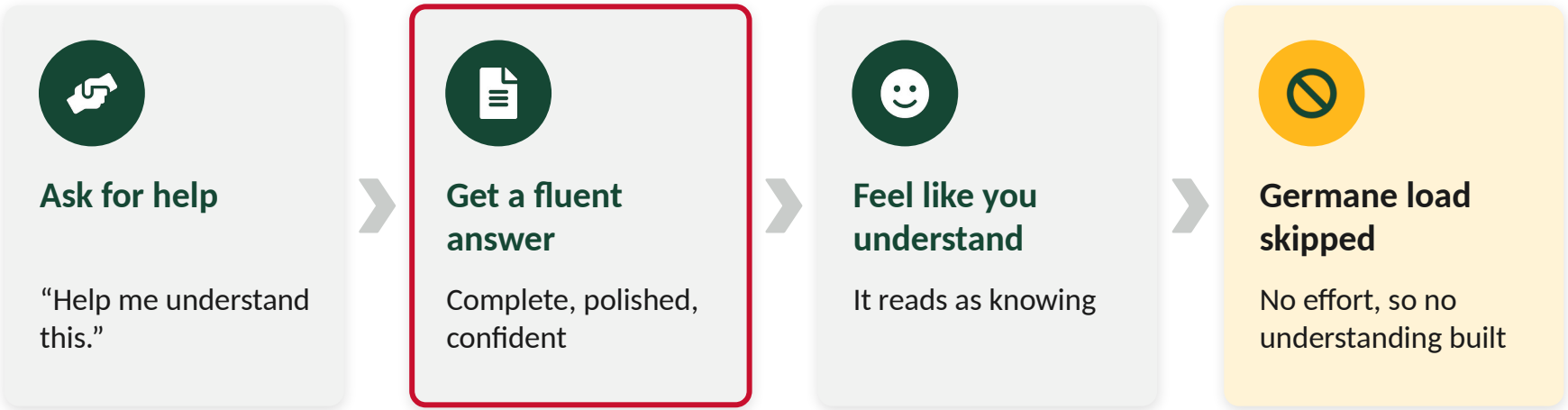
When used poorly,

AI can make learning fragile.

HELP

What happens when a student asks AI for help?

The illusion of learning



The illusion of learning is **fragile** learning.

Alstot (2026); Fisher et al. (2015); Klein & Klein (2025)

We don’t like pain! But …



learning is painful!

Fortified or fragile?

GERMANE LOAD Effort devoted to building understanding. Shaped by the learner.

Design for this!



Fragile

Chronic reliance on AI-generated answers bypasses the effortful processing that builds understanding.

Skips germane load



Fortified

Directing AI engagements to stimulate the effortful human processing that builds understanding.

Scaffolds germane load

Klein & Klein (2025); Sweller et al. (1998)

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Assignment Stress Test

1

List

What students do, step by step

The visible work, one task per line

2

Name

The thinking each step develops

Why they do it: the cognitive move

3

Run

The assignment through AI

As a student would. Note what AI produced.

4

Read

Each task on two axes

AI on its own. Students on their own.

Stress-test the thinking, not the deliverable.

Worked example: EDMS 523 Bundle 1

Lesson plan bundle on academic language, vocabulary, and English learners

What the candidate does	What AI produced on its own	AI
Describe the class context	Invented a plausible class. Fluent, and entirely fabricated.	Weak
Sequence the instruction	A full sequence, more detailed and connected than any pre-service candidate submission	Strong
Write the lesson rationale	Well reasoned rationale that accurately justified decisions made throughout the instructional sequence.	Strong
Connect decisions to theory	Strategies for ELL and students with accommodations tied to evidence-based pedagogy.	Strong
Strategy Book entry	Indistinguishable from a strong submission. Generated in seconds without any human direction required.	Strong

Start where students get stuck

You already know where students struggle on their own.

- They don't do well
- They get confused
- They get stuck

DECODING THE DISCIPLINES

Begin with a bottleneck: a point where many students' learning is interrupted on a concept or task.

Your experience as a disciplinary expert and teacher defines the student side of the stress test.

Middendorf & Pace (2004)

Worked example: EDMS 523 Bundle 1

Lesson plan bundle on academic language, vocabulary, and English learners

What the candidate does	What AI produced on its own	AI	Candidates
Describe the class context	Invented a plausible class. Fluent, and entirely fabricated.	Weak	Strong
Sequence the instruction	A full sequence, more detailed and connected than any pre-service candidate submission	Strong	Struggling
Write the lesson rationale	Well reasoned rationale that accurately justified decisions made throughout the instructional sequence.	Strong	Struggling
Connect decisions to theory	Strategies for ELL and students with accommodations tied to evidence-based pedagogy.	Strong	Struggling
Strategy Book entry	Indistinguishable from a strong submission. Generated in seconds without any human direction required.	Strong	Strong

Read each task on two axes

AI on its own **STRONG**

AI on its own **WEAK**

VULNERABLE

Where offloading can make learning fragile. Decide: integrate AI here? If so, in what role?

CUT OR COMPRESS

The task no longer builds or reveals the thinking.

TEACH

Human-essential, and a known struggle. Direct instruction, not an AI task.

PRESERVE

Students already do what AI cannot. Protect it.

Students on their own: **STRUGGLING**

Students on their own: **STRONG**

Worked example: EDMS 523 Bundle 1

Lesson plan bundle on academic language, vocabulary, and English learners

What the candidate does	What AI produced on its own	AI	Candidates	Read
Describe the class context	Invented a plausible class. Fluent, and entirely fabricated.	Weak	Strong	**PRESERVE**
Sequence the instruction	A full sequence, more detailed and connected than any pre-service candidate submission	Strong	Struggling	**VULNERABLE**
Write the lesson rationale	Well reasoned rationale that accurately justified decisions made throughout the instructional sequence.	Strong	Struggling	**VULNERABLE**
Connect decisions to theory	Strategies for ELL and students with accommodations tied to evidence-based pedagogy.	Strong	Struggling	**VULNERABLE**
Strategy Book entry	Indistinguishable from a strong submission. Generated in seconds without any human direction required.	Strong	Strong	**CUT**

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Design for this!

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AI Innovation

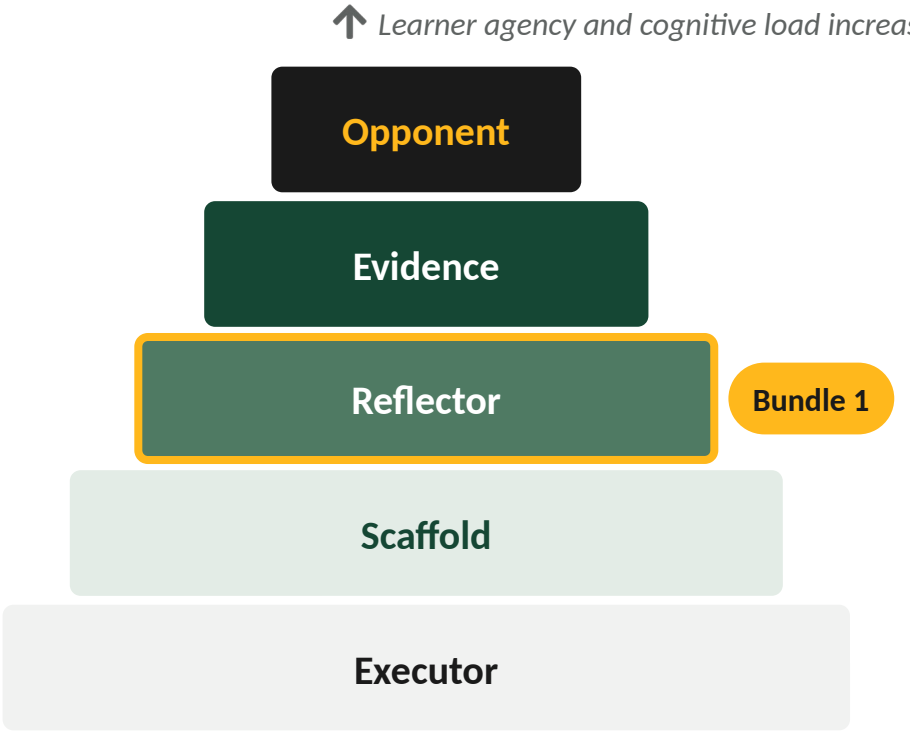
AI used to create novel approaches

WHY

Evidence of AI fluency (discernment and stewardship)

IWU AI Literacy Continuum (Bleistein & Snyder, 2026), adapted from the AIAS (Perkins et al., 2024)

Roles AI can take at the vulnerable point



AI as Opponent: Debates and tests the learner's assumptions

AI as Evidence: Its output becomes the object of critique

AI as Reflector: Makes the learner's own thinking visible

AI as Scaffold: Supports without deciding for the learner

AI as Executor: Performs the task on the learner's behalf

Adapted from Scott (2026), The AI Cognitive Use Pyramid, CC BY-NC-SA 4.0

The 3As and 4Ds in Instructional Design

Automation: AI checks spelling and grammar on finished work, disclosed. It writes nothing in the plan, the revisions, or the rationale.

Augmentation: In each reflector session, the candidate and AI think together: AI asks and reflects back; the candidate answers, decides, and revises.

Agency: I configure the coach, then let it run each session on my behalf with 28 candidates in 28 placements, within the limits I set.

Delegation: I handed AI **the reflective questions** I cannot ask 28 candidates in one week. The plan, revisions, rationale, and grading stay human.

Description: I described a master-teacher coach, one focus per session, the steps to follow in order, and an agenda built only from the candidate's own words.

Discernment: I stress-tested the bundle to find where AI is strong and candidates struggle (alignment, sequencing), and where AI is unreliable (standards).

Diligence: I wrote guardrails that hold AI in a Reflector role, like a professor's feedback conversation: it asks, never rewrites or grades, and leaves next steps to the candidate.

Framework: Dakan, Feller, & Anthropic (2025), CC BY-NC-SA 4.0. Applied to EDMS 523 Bundle 1.

From stress test to student interaction

1

Stress test

Find the vulnerability

2

Pick a role

Opponent, Evidence, Reflector, Scaffold, or Executor

3

Write the role

Use the templates provided

4

Test, refine, repeat

Test → refine → test → refine until you get the interaction you want students to have

5

Try it in class

Consider doing it in class

Next: my Bundle 1 coaching prompt, live

One assignment, redesigned

The original bundle



A lesson plan written once, then justified.

One assignment, redesigned

What I added: drafts, coaching, and reflection on the thinking



Cut: Teacher resource, Strategy Book entry

The lesson plan stopped being the evidence and became the material.

One assignment, redesigned

Where AI comes in

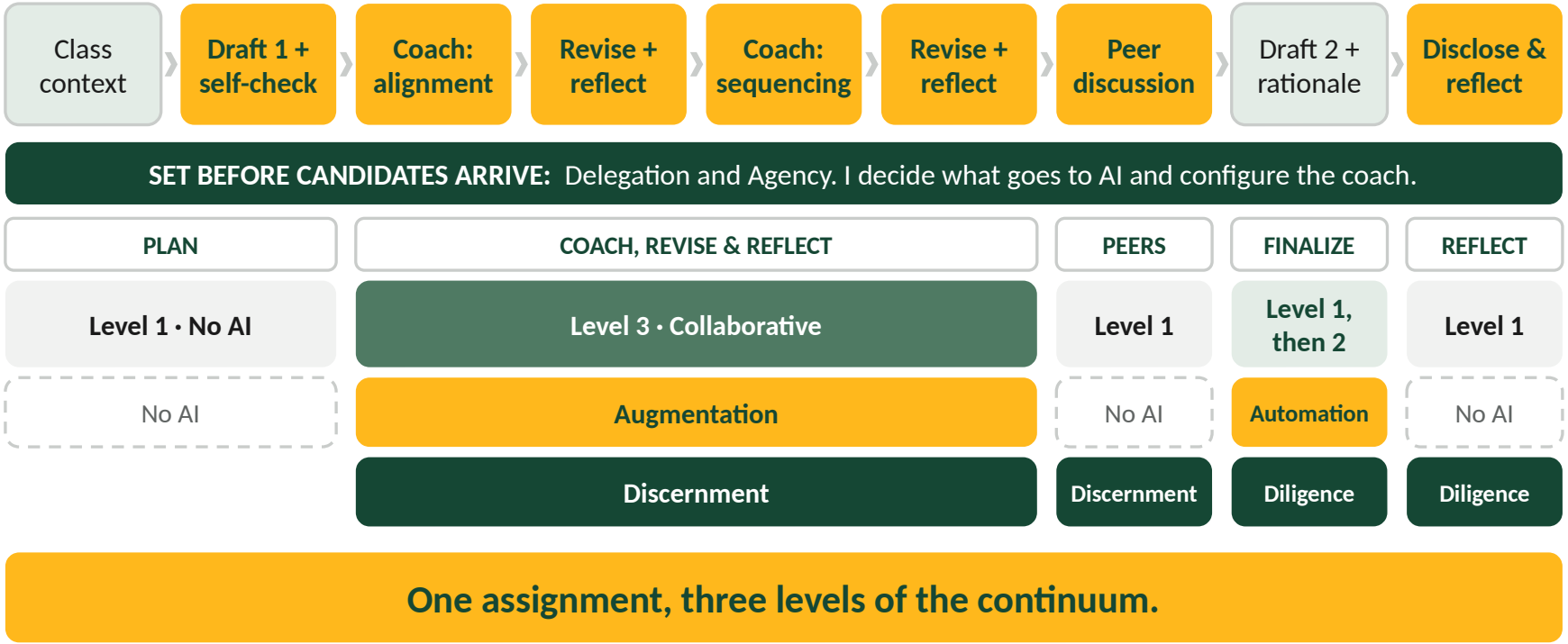
Class context › **Draft 1 + self-check** › **Coach: alignment** › **Revise + reflect** › **Coach: sequencing** › **Revise + reflect** › **Peer discussion** › Draft 2 + rationale › **Disclose & reflect**

PLAN ALONE	COACHED REVISION	PEERS	FINALIZE	REFLECT
No AI. The thinking starts with the candidate.	AI asks; the candidate answers, then revises alone and reflects on what changed.	No AI. Peers ask first.	No AI, except a spelling check.	No AI. Every use disclosed.

“Not to write your lesson. To ask you questions about it.”

One assignment, redesigned

Continuum levels · 3As (gold) · 4Ds (green): the assignment travels



One assignment, three levels of the continuum.

IWU AI Literacy Continuum (Bleistein & Snyder, 2026); Dakan, Feller, & Anthropic (2025), CC BY-NC-SA 4.0

PART 1 · HUMAN IN THE LOOP FOR LEARNING

In students' learning, AI asks. Students think.

NEXT · HUMAN IN THE LOOP FOR RESEARCH

The model assists. The researcher decides.

[Amy: full name, unit]

Work time · 20 minutes

OPTION A

Stress-test an assignment

1. Open Copilot or any chatbot
2. Open the Stress Test Guide and run the test
3. While it runs, work on What students do and Thinking it develops (columns B and C)

OPTION B

Build research guidance

1. Open Scopus AI
2. Open the Scopus AI Sandbox Checklist
3. Choose 2–3 prompts that align with your discipline from the sandbox checklist (or create your own). Enter them into Scopus AI and critically evaluate the generated Topic Summary, Concept Map, and Foundational Documents.

Debrief

1. What surprised you when AI ran your assignment?
2. Where is your assignment most vulnerable, and what will you decide there?
3. What is one next step you will take?

megan_alstot@baylor.edu · [Amy: email]

References

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